

Amitkumar Uttam Pandurnikar

B.Tech, Mechanical Engineering

SGGSIE&T, Nanded

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Education

2021-25	Shri Guru Gobind Singhji Institute of Engineering & Technology, Nanded	7.94/10
	Bachelor of Technology, Mechanical Engineering	
2021	12 th (Higher Secondary Examination)	77.17%
	State Board of Secondary Education	
2019	10 th (Secondary Examination)	53.6%
	CBSE Kendriya Vidyalaya SCR, Nanded	

Internships

5 th June -29 th July 2023	Inditech Valves Pvt. Ltd. , Pune Worked on a project involving the manufacturing of a fixture for an “ Integrated assembly, hydro testing, and calibration setup ” for industrial valves. Designed and developed the fixture to streamline assembly and testing processes, significantly reducing the required manufacturing and inspection time for valves while improving operational efficiency
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Academic Projects

Jan – May 2025	Parametric optimization in wire arc additive manufacturing (WAAM) of ER70S-6 using response surface methodology Achieved 90.5 % model accuracy, optimized best parameters for ER70S-6 and microstructural analysis by utilizing Scanning electron microscopy (SEM), Electron Backscatter Diffraction (EBSD), microhardness, Tensile testing, and Fractography in collaboration with IIT Bombay Microstructural Mechanics and Micro-forming (MMMMF) lab . Developed predictive models to analyze the influence of process parameters on bead geometry and material properties. Improved understanding of microstructure–property relationships to enhance the quality and performance of WAAM-fabricated components.
July – Dec 2024	Mechanical performance testing & Microstructural characterization of ER70S-6 low carbon alloy steel manufactured through wire arc additive manufacturing (WAAM) Evaluated mechanical performance and structural characteristics of ER70S-6 fabricated through wire arc additive manufacturing (WAAM), achieving improved hardness (236 BHN) compared to wrought mild steel (190 BHN). Identified distinct microstructural features such as equiaxed and columnar grain structures, contributing to a better understanding of structure–property relationships and supporting advancements in additive manufacturing for industrial applications.
11 th Mar – 6 th May 2024	ADAPTIVE STRUCTURE CREATION WITH THE HELP OF ARTIFICIAL INTELLIGENCE Applied AI-driven generative design in Autodesk Fusion 360 to create and optimize adaptive mechanical structures for improved strength, weight reduction, and design efficiency.
16 th Mar -22 Mar 2022	Automatic Street Light Controller Built an IoT-based automatic street light control system to reduce electricity wastage, where lights automatically turn ON when a vehicle is detected at a particular distance and turn OFF after it passes , improving energy efficiency and minimizing unnecessary power consumption.

Technical Skills & Interests

Feb – Jul 2024	Hands-on experience working with a Yaskawa 6-axis industrial robot, Wire Electric discharge machining (EDM), Optical Microscope, Manual and Semi-Automatic Polishing Machine, shaping machine, bandsaw machine, CNC milling, and lathe machine. 3D printer manager AICTE – SGGS IET IDEA Lab (Snap maker 350A model). Microstructure characterization, optimization methods, Response surface methodology, Taguchi Orthogonal Arrays (L8, L28, L18) , Fractography, CAD/CAM using Fusion 360, Product Lifecycle Management (PLM), Industry 4.0, Additive manufacturing
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